

AYUSH MADHUKAR

Bay Area, California | hello@ayumad.me | ayumad.me | github.com/ayumad

Computer Engineering student building practical AI and systems projects across local-first agents, knowledge retrieval, Linux infrastructure, and modern web interfaces. Focused on making complex personal technology reliable, understandable, and reusable.

EDUCATION

San José State University — B.S. Computer Engineering, in progress

Foothill College — Computer Science studies

SELECTED PROJECTS

Hermes | *Personal AI system*

- Designed a persistent agent backend on an M4 Mac mini so desktop, laptop, and mobile clients can share workflows and tools over a private mesh.
- Developing scoped memory and retrieval boundaries that surface useful context without exposing an unrestricted private notebook.

Obsidian RAG | *Knowledge retrieval*

- Building a visibility-aware retrieval layer for grounded question answering across a structured Obsidian vault.
- Preserves note metadata and source context so retrieval remains inspectable as models and embedding backends change.

Ayumad.me | *React, TypeScript, Vite, Vercel*

- Built a responsive portfolio and public knowledge layer with dynamic project, system, gear, and Markdown article routes.
- Created five content-aware visual modes with accessible controls, mode-specific motion, and optimized canvas particle rendering.

Homelab | *Proxmox, ZFS, virtualization*

- Repurposed a ThinkStation P520 as a virtualization and storage host for durable services and isolated experiments.
- Organizes workloads around clear recovery boundaries so testing can change without destabilizing core services.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, HTML/CSS

Systems: Linux, Proxmox VE, ZFS, Docker, virtualization, Tailscale

Web + AI: React, Vite, Vercel, retrieval-augmented generation, embeddings, agent workflows